**1** **(a)** $12x^{-3}$ or $\dfrac{12}{x^3}$ **[B1]**

**(b)** $\dfrac{2x^3}{3}$ or $\dfrac{2}{3}x^3$ **[B2]**

Correct cube root of coefficient $\dfrac{2}{3}$ OR correct index $x^3$ seen **[M1]**

*[Total Marks: 3]*

**2** **(a)** $20x$ **[B1]**

**(b)** Numerator $4x^8$ or denominator $y^6$ seen **[M1]**

$\dfrac{4x^8}{y^6}$ **[A1]**

**(c)** $-3$ **[B1]**

*[Total Marks: 4]*

**3** **(a)** $w^{11}$ **[B1]**

**(b)** $3^2$ or 9 seen, or $x^6$ seen **[M1]**

$9x^6$ **[A1]**

**(c)** $\dfrac{15}{5} = 3$ or $y^{8-(-2)}$ **[M1]**

$3y^{10}$ **[A1]**

*[Total Marks: 5]*

**4** **(a)** $p^{11}$ **[B1]**

**(b)** $9q^6$ **[B2]**

Partial correct answer $9q^k$ or $cq^6$ **[B1]**

**(c)** $(\sqrt[3]{27})^2$ or $3^2$ or $\sqrt[3]{729}$ **[M1]**

9 **[A1]**

*[Total Marks: 5]*

**5** **(a)** $\dfrac{1}{4}$ or 0.25 **[B2]**

Correct intermediate step e.g., $2^{-2}$ or $\left(\dfrac{1}{32}\right)^{\frac{2}{5}}$ or $32^{\frac{1}{5}} = 2$ **[M1]**

**(b)** Writes 125 as $5^3$ **[M1]**

Equates indices: $x + 2 = 3x$ **[M1]**

1 **[A1]**

*[Total Marks: 5]*

**6** **(a)** $\frac{1}{4}$ or 0.25 **[B1]**

**(b)** 25 or $\frac{1}{25}$ seen **[M1]**

1 **[A1]**

**(c)** $125 = 5^3$ seen or implied **[B1]**

$2x - 1 = 3x$ **[M1]**

$x = -1$ **[A1]**

*[Total Marks: 6]*

**7** **(a)** $4 \times \frac{3}{2} = 6$ or $x^{-3+\frac{1}{2}}$ **[M1]**

$6x^{-\frac{5}{2}}$ or $\frac{6}{x^{\frac{5}{2}}}$ **[A1]**

**(b)** $3^{2x+1} = 3^3$ or $2x + 1 = 3$ **[M1]**

1 **[A1]**

**(c)** $8^{\frac{1}{3}} = 2$ or $(a^6)^{\frac{1}{3}} = a^2$ or $(b^3)^{\frac{1}{3}} = b$ **[M1]**

$\frac{2a^2}{b}$ **[A1]**

*[Total Marks: 6]*

**8** **(a)** 8 **[B1]**

$x^9$ **[B1]**

$y^6$ as denominator or $y^{-6}$ **[B1]**

**(b)** $8 = 2^3$ seen or implied by $8^x = 2^{3x}$ **[M1]**

$x + 3 = 3x$ **[M1]**

1.5 or $\frac{3}{2}$ **[A1]**

*[Total Marks: 6]*

**9** **(a)** Expressing both sides with base 2: $(2^4)^{2x-1} = (2^3)^{x+2}$ **[M1]**

Equating powers: $4(2x - 1) = 3(x + 2)$ or $8x - 4 = 3x + 6$ **[M1]**

2 **[A1]**

**(b)** Taking the reciprocal to remove negative index: $\left(\frac{16y^8}{81x^{12}}\right)^{\frac{3}{4}}$ **[M1]**

Evaluating the fractional power of the numbers: $16^{\frac{3}{4}} = 8$ and $81^{\frac{3}{4}} = 27$ **[M1]**

$\frac{8y^6}{27x^9}$ **[A1]**

*[Total Marks: 6]*

**10** **(a)** $12x^{\frac{3}{2}}$ seen as numerator **[M1]**

$2x^k$ for any $k$ or $cx^{\frac{7}{6}}$ for any $c$ **[M1]**

$2x^{\frac{7}{6}}$ **[A1]**

**(b)** Converting all terms to base 3: $3^{2(2x-1)} \times 3^{x+1} = 3^{3x}$ **[M1]**

Applying exponent multiplication law: $3^{4x-2+x+1}$ or $3^{5x-1}$ **[M1]**

Equating indices: $5x - 1 = 3x$ **[M1]**

0.5 or $\frac{1}{2}$ **[A1]**

*[Total Marks: 7]*

**11** **(a)** $4x^2$ or $y^{-3}$ or $\frac{1}{y^3}$ seen **[M1]**

$\frac{4x^2}{y^3}$ or $4x^2y^{-3}$ **[A1]**

**(b)** $3^{2x}$ or $3^{3(x-1)}$ seen **[B1]**

$3\sqrt{3} = 3^{1.5}$ or $3^{\frac{3}{2}}$ seen **[B1]**

$2x + 3x - 3 = 1.5$ or equivalent equation in $x$ **[M1]**

$x = 0.9$ or $\frac{9}{10}$ **[A1]**

*[Total Marks: 6]*

**12** **(a)** $3x^2y^{-6}$ or $\frac{3x^2}{y^6}$ **[B3]**

Coefficient 3 seen **[M1]**

Correct subtraction of indices $x^{\frac{3}{2}-\left(-\frac{1}{2}\right)}$ leading to $x^2$ OR $y^{-2-4}$ leading to $y^{-6}$ **[M1]**

**(b)** Expresses all terms as powers of 2: $(2^2)^{2x-1} \times (2^3)^{x+1} = (2^5)^x$ **[M1]**

Correctly adds powers on the LHS: $2(2x-1) + 3(x+1)$ or $4x - 2 + 3x + 3$ **[M1]**

Forms linear equation equating LHS and RHS powers: $7x + 1 = 5x$ **[M1]**

$-\frac{1}{2}$ or $-0.5$ **[A1]**

*[Total Marks: 7]*